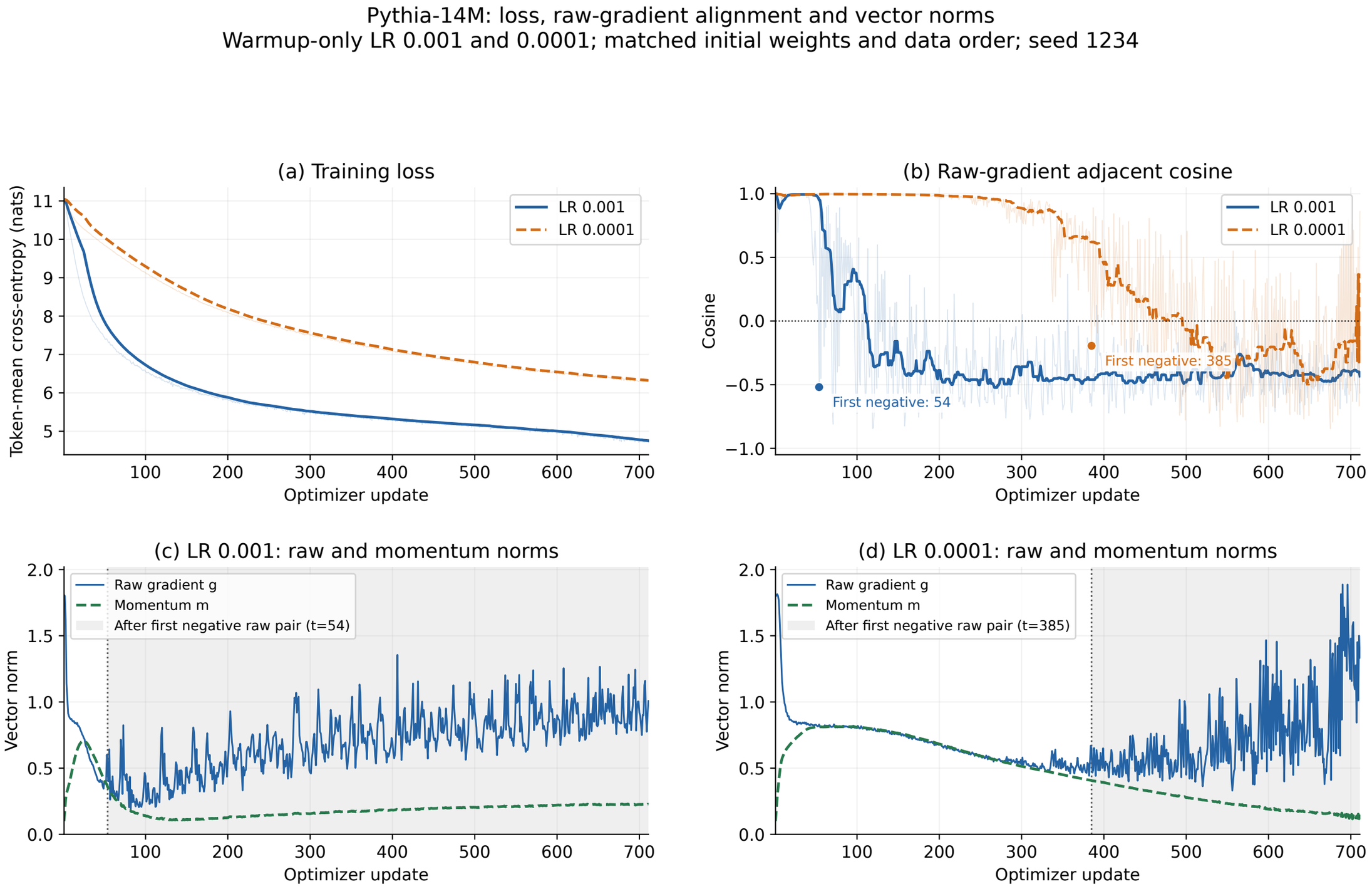

Pythia-14M: loss, raw-gradient alignment and vector norms
Warmup-only LR 0.001 and 0.0001; matched initial weights and data order; seed 1234
(a) Training loss
Token-mean cross-entropy (nats)
Optimizer update
LR 0.001
LR 0.0001
(b) Raw-gradient adjacent cosine
Cosine
Optimizer update
LR 0.001
LR 0.0001
First negative: 385
First negative: 54
(c) LR 0.001: raw and momentum norms
Vector norm
Optimizer update
Raw gradient g
Momentum m
After first negative raw pair (t=54)
(d) LR 0.0001: raw and momentum norms
Vector norm
Optimizer update
Raw gradient g
Momentum m
After first negative raw pair (t=385)
Light loss/cosine traces: individual updates. Bold loss: trailing 25-update token-weighted mean; bold cosine: trailing 25-update median.
Dots and vertical markers: first negative unsmoothed raw pair. Gray regions: time after that event, including later positive pairs.
Norm curves are unsmoothed on a shared linear scale. g is pre-clipping; m is uncorrected exp_avg from clipped gradients.
Display: full-sized updates 1-711; partial update 712 retained in data. Event timing describes an association; it does not define norm separation.